Question # 3 of 10 (Start time: 02:27:28 PM, 01 January 2026)

While solving the equation, 10x-(e^x)=0 by using Newton method, if x=0 is the initial approximate solution then the next approximation will be-----

Select the correct option

- ○ -1/10
- ○ 1/10
- ◉ 1/9
- ○ -1/9

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 2 of 10 (Start time: 02:27:10 PM, 01 January 2026)

For Rolle's Theorem, f is continuous on the interval _____.

Select the correct option

○	(a, b)
○	$(a, b]$
●	$[a, b]$
○	$[a, b)$

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 1 of 10 (Start time: 02:26:32 PM, 01 January 2026)

$1^2+2^2+3^2+...+15^2$ equals _____.

Select the correct option

○	15(16)(30) / 6
●	15(16)(31) / 6
○	15(16)(29) / 6
○	15(16)(17) / 6

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 10 of 10 (Start time: 02:29:26 PM, 01 January 2026)

By applying mean value theorem to the function
f(x)=lnx in the interval [1,e], the corresponding value of 'c' is ----------(Hint: lne=1, ln1=0)

Select the correct option

○	-e
○	e
○	1-e
◉	e-1

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 9 of 10 (Start time: 02:29:04 PM, 01 January 2026)

Which of the following will be left end points if the interval [-2,2] is divided into 4 equal subintervals.

Select the correct option

●	-2,-1,0,1
○	-2,-1,1,2
○	None of these
○	-1,0,1,2

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 8 of 10 (Start time: 02:28:49 PM, 01 January 2026)

Integration of (Cosx/Sinx).Cosecx with respect to x.....

Select the correct option

○	Cotx
○	Cosecx
◉	-Cosecx
○	Secx

MTH101 - Calculus And Analytical Geometry (Quiz No. 3) **Quiz Start Time: 02:26 PM, 01 January 2026**

Question # 7 of 10 (Start time: 02:28:27 PM, 01 January 2026) **Total Marks:** 1

In Rolle's theorem, f(x) is continuous in closed interval [a,b] and differentiable in open interval (a,b).Then why we don't discuss its differentiability in the closed interval [a,b] because ----------

Select the correct option

- f '(a)=f '(b)=0
- f(a) and f(b) never exist
- f '(a) and f '(b) never exist
- f(a)=f(b)=0

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 6 of 10 (Start time: 02:28:16 PM, 01 January 2026)

The symbol $\int$ was introduced by_____and is called integral sign.

Select the correct option

○	Cauchy
○	Newton
○	Lagrange
◉	Leibnitz

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 5 of 10 (Start time: 02:28:02 PM, 01 January 2026)

Integration of 4Cosx with respect to x is.....

Select the correct option

- -4Sinx
- 4Sinx

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 4 of 10 (Start time: 02:27:45 PM, 01 January 2026)

How many subintervals of length '2' will be formed for the interval [4,16] ?

Select the correct option

○	5
○	4
○	7
◉	6